Abdallah Ahmed

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Note: *As requested in the form, this document provides the direct link and an overview of my engineering portfolio repository (instead of a CV).*

Portfolio GitHub Repository

https://github.com/3bdallahhahmed/Abdalla-Ahmed-portfolio

Contains project documentation, CAD files, Altium PCB files, photos, and demo videos for all my work.

Scan with phone

Projects Included in the Repository

1. Single-Actuator Hexapod Walking Robot — E-JUST • MTR 221 Theory of Machines

Designed and built a six-legged robot driven by a single motor using planar linkages and an internal 90° bevel gearbox. Modeled in SolidWorks and 3D printed with tuned pin tolerances.

`Folder: 01-hexapod-walking-robot/ (CAD renders, gear cutaways, trajectory, walking demo video)`

2. Robotics & Computer Vision Summer Track — Electronics Research Institute (ERI)

60-hour intensive summer training at ERI. Programmed ABB industrial arms (IRB 1200 and YuMi collaborative robot) in RAPID, and ran deep learning vision tasks (DeepLabV3 vs FCN semantic segmentation, YOLOv8 pose estimation).

`Folder: 02-eri-robotics-simulation/ (Photos of ABB robots, segmentation tests, pose detection, lab video)`

3. Satellite PCB Design Workshop — Egyptian Space Agency (EgSA)

60-hour hardware training in Altium Designer. Designed an Arduino Uno R3 compatible board (custom schematic, IPC footprints, 2-layer routing) and a 5V regulated linear power supply.

`Folder: 03-egsa-pcb-design/ (3D board models, routing layouts, full schematics)`

4. Autonomous Air Quality Monitoring Segway Robot — SoiLAB Eco-Warrior Camp • UAE

Two-wheeled self-balancing robot with discrete PID control, 6-axis IMU sensor fusion, and real-time environmental particulate (PM2.5/PM10) and gas sampling.

`Folder: 04-soilab-air-quality-robot/ (Balancing dynamics, control notes, hardware photos)`

5. Etba3ly 3D Printing Prototyping — Maker Project • etba3ly.store

Student-run rapid 3D printing service providing custom prototyping, functional replacement parts, and scale display models (e.g. F-35 Lightning II model).

`Folder: 05-etba3ly-3d-printing/ (Scale model print, service details)`